
DATA WRANGLING COMPREHENSIVE STUDY GUIDE

FIT5196 - Complete Concept Map (REVISED WITH ALL VISUAL CONTENT)

PART 1: FOUNDATIONS & FUNDAMENTALS

What Is Data Wrangling?

Data Wrangling is the process of acquiring, cleaning, structuring, and enriching raw data into a format that is directly usable for analysis.

Beginner Hook: Think of data wrangling like preparing ingredients for cooking. Raw ingredients (messy data) need washing, chopping, organizing, and combining before you can create a meal (conduct analysis). You wouldn't throw unwashed produce into a blender—similarly, you can't analyze raw, messy data effectively.

Quiz Alert: Know this definition verbatim—it appears consistently across all weeks. The university emphasizes that data wrangling is about making data "directly usable for analysis."

Why Data Wrangling Matters

Data wrangling is a **preparatory step for data analysis** that is **essential for ensuring that data analysis leads to accurate and actionable insights**.

The critical fact: **Data scientists spend 60% of their time on cleaning and organizing data**, compared to only 19% collecting data, 9% mining for patterns, 4% refining algorithms, 3% building training sets, and 5% on other tasks.

Goals of Data Wrangling:

- **Improving Data Quality:** Ensuring accuracy and completeness
- **Data Formatting and Standardization:** Making data consistent
- **Simplifying Access to Data:** Making data retrievable and understandable
- **Enriching Data:** Adding context and additional information
- **Reducing Data Complexity:** Simplifying structure for analysis
- **Facilitating Data Integration:** Combining multiple sources
- **Increasing Analytical Efficiency:** Speeding up the analysis process

- **Supporting Decision Making:** Enabling better business intelligence

Challenges in Data Wrangling

Quiz Alert: This is a heavily tested section. Know all eight challenges.

Challenge	What It Means	Example
Volume of Data & Scalability	Handling massive datasets that grow faster than processing capacity	3+ billion Facebook users monthly; processing billion-row datasets
Data Quality Issues	Errors, inconsistencies, and inaccuracies in raw data	Manual entry errors, measurement device malfunction, missing values
Data from Diverse Sources	Data comes in different formats and structures	JSON, XML, CSV, Excel, PDF, PNG, JPEG files all in one project
Complexity of Data Structures	Dealing with unstructured, semi-structured, and highly nested data	HTML documents, logs, images, hierarchical JSON
Lack of Standardization & Interpretability	No agreed-upon format or naming conventions	"Male" vs "M", "01/12/2024" vs "01-12-2024", different currencies
Highly Time-Consuming	Data preparation takes significant effort	60% of data scientist's time spent on this phase
Skill and Tool Requirements	Need expertise across multiple domains	Python, R, SQL, statistics, NLP, databases, business knowledge
Data Privacy and Security	Protecting sensitive information and complying with regulations	GDPR, HIPAA compliance; encryption requirements

Beginner Hook: Imagine you're combining recipes from ten different countries. Some use cups, some use grams. Some call salt "sal," some "sodium chloride." Before you can follow any recipe, you need to standardize everything—that's what data wrangling does.

PART 2: THE DATA WRANGLING PROCESS

Overview: Eight Core Tasks

The Data Wrangling Process Flow:

Data Discovery → Data Collection → Data Cleaning → Data Pre-processing
↓

Data Transformation → Data Enrichment → Data Storing → Data Validation

Quiz Alert: Memorize this sequence. The university emphasizes this specific order consistently.

Each stage builds on the previous one. You cannot effectively clean data you haven't discovered; you cannot transform data you haven't collected.

Task 1: Data Discovery

Data Discovery is the initial step where data scientists identify and access various data sources relevant to the analytical objectives, including determining availability, suitability, and quality.

Data Discovery Process Components:

A. Identification of Data Sources

There are two categories:

Internal Sources (Secondary Data Already Owned):

- Business records
- Historical data
- Transactional data

External Sources:

- Government publications
- Academic and research institutions
- Industry reports
- Public databases and archives
- Media archives
- Social media and online content

Secondary Data vs. Primary Data:

Aspect	Secondary Data	Primary Data
Origin	Collected by someone else for different purpose	Collected firsthand by researcher for specific purpose
Control	Limited control over quality/methodology	Full control over collection process
Cost	Cost and time efficient	Expensive and time-consuming
Access	Access to broad, diverse data	Specific to research question

Quality	May be outdated or of uncertain quality	Rigorously collected to specifications
Relevance	May not perfectly match current needs	Perfectly aligned with current analysis

Beginner Hook: Secondary data is like buying pre-made ingredients from a store (someone else prepared them); primary data is growing your own vegetables (you control everything).

Healthcare Analytics Example: Patient Readmission Prediction

Project Objective: Develop a predictive model to identify patients at high risk of readmission within 30 days of discharge.

Project Goal: Improve patient outcomes and reduce healthcare costs.

Data Sources Identified:

From Hospital Systems:

- Electronic Health Records (EDA): Patient diagnoses, treatments, medications, outcomes
- Hospital Admission, Discharge & Transfer Systems: Dates and reasons for movements within hospital
- Insurance Claim Data: Claims, procedures performed, billing items, reimbursement rates

From External Sources:

- Wearable Device Data: Vital signs (heart rate, activity levels)
- Treatment Plans: Discharge medications, follow-up appointments

Data Accessibility:

- Hospital IT Department manages databases
- Clinicians provide medical expertise
- Insurance companies hold claim data
- Wearable companies provide device data streams

Data Quality Assessment Required:

- **Completeness:** Are all patient records present?
- **Accuracy:** Do diagnoses match reality?
- **Formatting:** Are dates consistent? Patient IDs matching?
- **Coverage:** Do we have multi-year data? All departments?

B. Data Profiling and Assessment

Three components:

Understanding Data Structure:

The **universe of data structures** you'll encounter:

1. **Structured Data** - Highly organized, easily understood by machines
 - Relational databases (SQL databases with tables, rows, columns)
 - Data warehouses
 - Tables with clear schemas
 - **Example:** Patient database with columns: Patient_ID, Name, DOB, Diagnosis
2. **Unstructured Data** - No pre-defined model, challenging to process
 - Text data (clinical notes, reports)
 - Multimedia data (images, video, audio)
 - Social media posts
 - Email content
 - **Example:** Doctor's handwritten notes converted to text; CT scan images
3. **Semi-Structured Data** - Has some organization but not fully structured
 - JSON (JavaScript Object Notation) - hierarchical key-value pairs
 - XML (Extensible Markup Language) - nested tags
 - Logs from systems
 - Sensor data
 - **Example:** Patient data sent via API in JSON format with nested medications array
4. **Time-Series Data** - Sequences of values indexed in time order
 - Regular intervals (e.g., daily stock prices, hourly sensor readings)
 - Irregular intervals (e.g., event logs when events occur)
 - Used in financial analysis, sensor data, performance monitoring
 - **Example:** Patient vitals: heart rate measurements every 5 minutes during hospital stay
5. **Graph Data** - Models relationships between entities flexibly
 - Social network analysis (connections between people)
 - Recommendation systems (products related to other products)
 - Fraud detection (transactions connected through accounts)
 - **Example:** Doctor-patient relationships, medication interactions
6. **Big Data** - Data so voluminous traditional processing can't manage it
 - Encompasses all above structures but at massive scale
 - Characterized by Volume, Velocity, and Variety (the 3 Vs)
 - **Example:** Real-time monitoring of 1 million patients' vital signs

Beginner Hook: Structured data is like an organized filing cabinet (you know exactly where everything is); unstructured data is like a pile of loose papers (you have to dig to find what you need).

Content Exploration - Understanding what information the data holds (categorical, numerical, textual)

Quality Assessment - Identifying issues: missing values, duplicates, inconsistencies

C. Data Cataloging and Metadata Management

Metadata is information about the data itself. **Quiz Alert:** Know the six types:

Metadata Type	What It Captures	Example
Descriptive Metadata	Information about data content	Title, author, keywords, subject
Structural Metadata	How data is organized	How files relate, storage format
Technical Metadata	Technical details	File format, encoding, software used
Administrative Metadata	Who manages data, access rights	Owner, creation date, access permissions
Provenance Metadata	Data origins and transformations	Where it came from, how it was modified
Usage Metadata	How data is used	Access logs, usage statistics, applications

Key Activities:

- **Catalog Creation:** Make data discoverable and understandable
 - **Data Lineage Documentation:** Trace data from source through all transformations to current state
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Task 2: Data Collection

Data Collection is the systematic process of gathering and measuring information on variables of interest in an established systematic fashion enabling research questions to be answered, hypotheses tested, outcomes evaluated.

Considerations for Data Collection:

When designing data collection, address these twelve factors:

1. **Define Objective** - What do you want to learn?
2. **Data Requirements** - What specific data meets the objective?
3. **Data Sources** - Where to obtain the data?
4. **Data Quality** - Ensure high quality at collection point
5. **Data Volume** - Assess volume needed and processing capacity
6. **Ethics** - Comply with legal standards and permissions
7. **Data Format** - What format to use?
8. **Collection Methods & Tools** - Choose appropriate methods and tools
9. **Timeframe and Budget** - Align with project deadlines and budget
10. **Data Privacy and Security** - Protect personal information
11. **Documentation** - Keep detailed records of collection process

12. Pilot Test - Validate process and instruments before full rollout

Quantitative vs. Qualitative Data:

Aspect	Quantitative Data	Qualitative Data
Nature	Numerical, measurable	Descriptive, conceptual
What It Shows	Quantity and magnitude	Qualities and characteristics
Collection	Surveys, experiments, structured instruments	Interviews, observation, open-ended questions
Analysis	Statistical analysis, hypothesis testing	Interpretive, thematic analysis
Display	Graphs, charts, tables	Text, transcripts, narratives
Purpose	Formulate facts, uncover patterns	Understand concepts, develop hypotheses

Task 3: Data Pre-Processing

Data Pre-Processing is preliminary data preparation where raw data is made more suitable for analysis through selection and formatting.

Selection Methods:

A. Subsetting

- **Definition:** Choosing a subset of data relevant to analysis by filtering rows or columns based on criteria
- **Example:** From airline crash data spanning 1920-2020, subset only 2000-2010 accidents for analysis
- **Use Case:** Focusing analysis on specific time period, geographic region, or category

B. Sampling

- **Definition:** Drawing a representative sample when dataset is too large or for exploratory analysis
- **Example:** Analyzing Facebook's 3.05 billion monthly users by sampling 3.05 million users (0.1% of data)
- **Facebook User Breakdown (from 2023 data):**
 - India: 385.65 million users (12.64%)
 - United States: 188.6 million users (6.18%)
 - Indonesia: 136.35 million users (4.47%)
 - Brazil: 111.75 million users (3.66%)
 - Mexico: 94.8 million users (3.11%)
- **Statistical Purpose:** Reduces computational burden while maintaining representativeness

Beginner Hook: Sampling is like tasting a spoonful of soup to check seasoning instead of tasting the entire pot.

Formatting Methods:

Quiz Alert: Know these formatting types—they're universally tested.

Data Type	Formatting Options	Why It Matters
Date Formatting	DD/MM/YYYY, MM/DD/YYYY, YYYY/MM/DD	Ambiguous dates (01/02/2024) mean different things in different locales
Numeric Formatting	Decimal points (. vs ,), thousands separators (1,000 vs 1.000)	Calculations fail if format inconsistent
Text Formatting	Upper, lower, title case; whitespace; symbols	"Data Scientist" vs "data scientist" vs "data scientist"
Categorical	'Male' vs 'M', 'Female' vs 'F'	Must standardize for grouping and analysis
Currency	USD, AUD, EUR, CNY conversion	Need consistent currency or conversion rates
Unit Conversion	Miles→Km, Fahrenheit→Celsius, inches→cm	Impossible to analyze if units differ
Text Encoding	UTF-8 standardization	Prevents garbled characters and data loss
Time Zone	Align to single timezone	Critical for time-series analysis
Address Formatting	Standardize street, city, postcode	Essential for location-based analysis
File Format	XML→CSV, JSON→XML	Enable compatibility with tools

Task 4: Data Cleaning

Data Cleaning is a critical step focused on removing or correcting inaccuracies, inconsistencies, duplicates, or irrelevant data points within datasets to ensure quality and reliability.

Data Anomalies:

Anomalies are items that deviate from typical patterns. They can indicate errors OR reveal valuable insights.

Four Types:

Anomaly Type	Definition	Example	Action
Missing Values	Data points absent from records	Customer phone number not recorded	Impute, remove row, or flag

Outliers	Extreme values outside expected range	Patient weight: 500kg in dataset of 50-150kg people	Investigate, remove, or cap
Duplicate Data	Identical records appearing multiple times	Same customer in database twice with slightly different names	Identify, merge, remove duplicates
Consistency Issues	Same entity represented differently across sources	Customer named "Bob Smith" in one system, "Robert Smith" in another	Standardize to single representation

Beginner Hook: Data cleaning is like going through your refrigerator: throw out expired food (missing values), separate the rotten items (outliers), consolidate duplicates, and organize everything consistently.

Task 5: Data Transformation

Data Transformation involves **converting data from original form into format/structure more appropriate for analysis** through normalization, aggregation, feature engineering.

Numeric Data Transformations:

A. Normalization

Definition: Rescales numeric data to fall within specified range (typically 0-1) so no variable dominates due to scale.

Formula:

$$\text{Normalized Value} = (X - \text{Min}) / (\text{Max} - \text{Min})$$

Example: If salary ranges \$20,000-\$120,000:

- \$20,000 → 0
- \$120,000 → 1
- \$70,000 → 0.625

Why It Matters: Machine learning algorithms perform better when features are on similar scales.

B. Standardization

Definition: Adjusts data so it has mean of 0 and standard deviation of 1.

Formula:

$$\text{Z-Score} = (X - \text{Mean}) / \text{Standard Deviation}$$

When to Use: When algorithms assume normally distributed data (e.g., linear regression, neural networks).

Beginner Hook: Normalization puts data on a 0-1 scale (like percentage); standardization centers it around zero (like temperature variations around freezing point).

C. Aggregation

Definition: Summarizing or combining data by computing summary statistics for groups.

Examples:

- Mean, median, sum of values
- Daily sales → Monthly total sales
- Individual test scores → Class average

Purpose: Reduces dimensionality, extracts insights from granular data.

D. Discretisation

Definition: Converts continuous variables into discrete by creating equal-width or equal-frequency intervals.

Example: Age (continuous 0-100) → Age Groups (discrete: 0-18, 19-35, 36-50, 50+)

Use Case: Some algorithms work better with categorical data; easier interpretation.

E. Binning/Bucketing

Definition: Groups numeric values into bins or categories.

Example:

- Income (\$0-\$30k, \$30k-\$60k, \$60k-\$100k, \$100k+)
- Student scores (A: 90-100, B: 80-89, C: 70-79)

Outcome: Simplifies model, helps identify trends.

Text Data Transformations:

A. Tokenization

Definition: Breaking text into individual units (tokens)—typically words, phrases, or symbols.

Types:

- **Word Tokenization:** "I love data" → ["I", "love", "data"]
- **Sentence Tokenization:** Paragraph split by sentences
- **Subword Tokenization:** "unbelievable" → ["un", "believ", "able"]
- **Character Tokenization:** "data" → ["d", "a", "t", "a"]

Why Essential: First step for NLP tasks; enables analysis of text components.

B. Stemming

Definition: Reduces words to root form by removing prefixes/suffixes/pluralization.

Examples:

- watches → watch
- parties → party
- carrying → carry

Result: Multiple word forms map to single stem.

Limitation: May produce non-words; doesn't consider context.

C. Lemmatization

Definition: Advanced form of stemming using context, vocabulary, morphological analysis, and grammatical information to determine lemma (dictionary form).

Examples:

- "better" → "good" (uses context to understand the word means)
- "was" → "be" (identifies verb conjugation)

Why Better: Produces actual words; considers grammar and meaning.

Beginner Hook: Stemming is crude cutting (watches→watch); lemmatization is intelligent reduction (was→be, better→good, using grammar rules).

D. Vectorization - Count Vectorizer (Key Method)

Definition: Converting text into numerical format for computational analysis.

Three Steps:

1. **Tokenization:** Text split into tokens
2. **Vocabulary Building:** Create list of unique tokens across corpus
3. **Count Calculation:** Count word frequency per document

Example:

Document 1: "The cat sat on the mat" Document 2: "The dog sat on the log"

After tokenization, vocabulary: ["the", "cat", "sat", "on", "mat", "dog", "log"]

Document-Term Matrix:

	the	cat	sat	on	mat	dog	log
Doc1	2	1	1	1	1	0	0
Doc2	2	0	1	1	0	1	1

Purpose: Converts unstructured text to structured numerical data for algorithms.

Image Data Transformations:

A. Geometric Transformations:

- **Resize:** Change dimensions
- **Rotation:** Turn image by degrees
- **Flipping:** Mirror horizontally/vertically
- **Translation:** Shift image position
- **Cropping:** Extract portion of image

B. Color/Pixel Transformations:

- **Normalization:** Scale pixel values (0-255) to (0-1) or (-1 to 1)
- **Color Space Conversion:** RGB → Grayscale/Black & White, RGB → CMYK
- **Brightness Adjustment:** Increase/decrease luminosity
- **Contrast Adjustment:** Enhance difference between light and dark areas

C. Enhancement Transformations:

- **Noise Injection:** Add random noise for data augmentation
- **Filtering:** Apply blur, sharpening, edge detection

Real-World Example - Lena Image Processing: The famous "Lena" image (from 1972 Playboy Magazine) demonstrates:

- **Original RGB Color:** Full color photo
- **Grayscale Conversion:** Black and white version

- **Edge Detection:** Highlights boundaries and contours
- **Other Filters:** Blur, sharpening, texture extraction

Beginner Hook: Image transformation is like photo editing—resize (crop), color correct (normalize), enhance (contrast), and add effects (noise).

Task 6: Data Enrichment

Data Enrichment involves **enhancing data with additional context or merging with other sources** to make it more complete and informative.

Four Methods:

Method	What It Does	Example
Data Integration	Combining data from different sources into one dataset	Merging customer CRM data with transaction data
Data Augmentation	Adding information to existing records to enhance depth	Adding demographic data to customer records
Attribute Enrichment	Adding new features/attributes not previously included	Adding "age group" or "customer lifetime value" to customer records
Temporal Enrichment	Adding time-related data	Adding "day of week", "season", "fiscal quarter"
Semantic Enrichment	Adding metadata or semantic information for understanding	Adding data dictionary definitions, business context

Beginner Hook: Enrichment is like adding ingredients to a basic recipe to make it richer and more complete.

Task 7: Data Storing

Data Storing involves **saving wrangled data in suitable storage system** for easy access and analysis—databases, warehouses, cloud storage.

Key Considerations:

1. **Selection of Storage Solution** - Choose system matching data nature, size, intended use
2. **Data Modelling and Schema Design** - Design logical structure supporting efficient storage/query
3. **Normalization and Denormalization** - Reduce redundancy (relational) OR optimize performance (NoSQL)
4. **Data Formatting and Encoding** - Ensure consistent format matching system requirements

5. **Implementing Security Measures** - Protect from unauthorized access; ensure GDPR/HIPAA compliance
 6. **Data Indexing and Optimization** - Enhance speed/efficiency of retrieval
 7. **Backup and Recovery Planning** - Ensure durability and recovery capability
 8. **Data Lifecycle Management** - Manage from creation to deletion per retention policies
 9. **Documentation and Metadata** - Provide context for understanding and governance
 10. **Monitoring and Maintenance** - Keep system efficient, scalable, reliable as data grows
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Task 8: Data Validation

Data Validation ensures **data is accurate, consistent, and usable for analysis** through checks verifying quality and correctness.

Validation Techniques:

Technique	Purpose	How It Works
Define Validation Rules	Establish standards data must meet	Set acceptable ranges, formats, relationships
Accuracy Check	Ensure data reflects real-world values	Verify customer age is realistic, prices are current
Consistency Check	Verify data consistency within/across datasets	Check that status codes match across systems
Completeness Validation	Ensure critical data isn't missing	Flag records missing required fields
Logical Integrity Test	Confirm data makes logical sense	If start_date, then end_date must be later
Range and Constraint Validation	Ensure values fall within acceptable ranges	Age 0-150, temperature -50°C to 50°C
Format Validation	Verify expected format/structure	Email matches pattern, date is valid
Uniqueness Check	Ensure entries supposed to be unique aren't duplicated	Primary key violations, duplicate emails
Cross-Validation	Validate using related datasets or sources	Compare customer info across CRM and billing systems
Automate Validation	Streamline for efficiency and repeatability	Automated scripts for large/ongoing datasets
Document Issues	Track identified issues and resolutions	Maintain audit trail for accountability

Quiz Alert: Know this complete list—validation is a heavily tested concept.

PART 3: UNDERSTANDING DATA TYPES & EDA

Common Data Types

Identifying data types during EDA is **fundamental** because it dictates analysis methods, visualizations, preprocessing, and model selection.

Type 1: Numerical Data

Definition: Quantifiable information expressed using numbers.

Two Subtypes:

A. Continuous Data

- **Definition:** Any value within a range; divisible infinitely
- **Examples:** Height, weight, temperature, speed, distance
- **Analysis:** Mean, median, standard deviation; scatter plots; regression
- **Visualization:** Histograms, boxplots, line charts

B. Discrete Data

- **Definition:** Countable values; whole numbers only
- **Examples:** Student attendance, parking bays available, lines of code, dataset size
- **Analysis:** Frequency counts, mode
- **Visualization:** Bar charts, count plots

Visual Representation: Continuous and Discrete data shown in Venn diagram:

- **Continuous** (Blue circle): "Any value within a range and can be divided infinitely" (Height, Weight, Temperature, Speed)
- **Discrete** (Green circle): "Countable values or categories" (Student attendance, Available parking bays, Number of coding lines, Size of a data set)

Beginner Hook: Continuous is smooth (a person's height changes gradually); discrete is stepped (you either attended class or you didn't).

Type 2: Character Data

Definition: Single alphabetic letters, symbols, or numerals (not as numbers) stored in fixed-length format.

Examples: 'A', 'b', '1', '6', '?', '!', '#'

Properties: Individual units; basic building block of text

ASCII Table Reference: Character data maps to decimal and hexadecimal values:

- 'A' = Decimal 65, Hex 41
- 'a' = Decimal 97, Hex 61
- '0' = Decimal 48, Hex 30
- Space = Decimal 32, Hex 20
- Control characters: NULL (0), START OF HEADING (1), etc.

Important Note: Character data includes single alphabetic letters, symbols, and numerals that do NOT have numeric value. It's the basic unit used to construct strings (text) and is typically stored in a fixed-length format.

Type 3: Text Data

Definition: Strings (sequences of characters)—words, sentences, paragraphs, documents.

Properties:

- Requires special handling for NLP
- Demands encoding or tokenization
- Used for qualitative analysis
- Often requires transformation before computational use

Examples: Customer reviews, social media posts, email content, survey responses

Type 4: Categorical Data (Implied throughout, given context about encoding)

Definition: Data with fixed, finite set of categories.

Examples:

- Binary: Yes/No, Male/Female, True/False
- Multi-class: Color (Red, Blue, Green), Region (North, South, East, West)

Type 5: Image Data

Definition: Visual information encoded as bitmap—set of pixel values.

Properties:

- Used in computer vision, image recognition
- Processing involves CNNs (Convolutional Neural Networks)
- Requires transformations: resize, color conversion, normalization

Real-World Example - Lena Image Processing: The "Lena" image demonstrates image transformation techniques:

- **Original:** Full RGB color image
- **Grayscale:** Converted to black and white
- **Edge Detection:** Highlights image contours and boundaries
- **Other Transformations:** Filters, textures, compression

Example: Medical imaging, satellite photos, facial recognition datasets

Why Understanding Data Types Matters

Quiz Alert: Know all six reasons—test format often asks "why identify data types in EDA?"

Reason	Impact	Consequence If Wrong
Appropriate Analysis	Different types need different statistical methods	Using categorical tests on continuous data gives meaningless results
Effective Visualization	Each type visualizes differently optimally	Plotting categorical as scatter plot = uninterpretable mess
Correct Preprocessing	Different types need different cleaning techniques	Applying normalization to categorical data = data corruption
Effective Feature Engineering	Data type determines what features can be created	Can't take square root of categorical variable
Suitable Algorithms	Some algorithms require specific data types	Decision trees struggle with unencoded categorical data
Data Integrity	Operations must be valid for the type	Arithmetic operations on categorical data = invalid conclusions

Exploratory Data Analysis (EDA) for Numerical Data

EDA Definition: Crucial step using statistical graphics, plots, and information tables to explore and summarize main characteristics of data without making assumptions about contents.

Purpose: Identify underlying structure, detect outliers, test assumptions, develop initial ideas for models.

Descriptive Statistics: Central Tendency

A. Mean (Average)

Formula:

$$\mu = (x_1 + x_2 + \dots + x_n) / n$$

Example: Dataset [4, 8, 6, 5, 3]

$$\mu = (4 + 8 + 6 + 5 + 3) / 5 = 26/5 = 5.2$$

Interpretation: On average, values are 5.2

B. Median (Middle Value)

Definition: Middle value when all values arranged in order.

For Odd Number of Values:

Dataset [3, 4, 5, 6, 8] → Median = 5

For Even Number of Values:

Dataset [3, 4, 5, 6] → Median = $(4 + 5) / 2 = 4.5$

When to Use: More robust to outliers than mean; better for skewed distributions.

C. Mode (Most Frequent)

Definition: Most frequently occurring value.

Example: Dataset [2, 3, 4, 4, 5, 5, 5, 6]

Mode = 5 (appears 3 times)

When to Use: Best for categorical data; useful for identifying typical value when distribution has clear peak.

Quiz Alert: Class question asks "which is best for EDA on new dataset?" Answer: **All three should be examined** to understand complete picture. Mode gives most common, median shows typical resistant to outliers, mean shows average.

Descriptive Statistics: Dispersion (Spread)

A. Range

Formula:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example: Dataset [4, 8, 6, 5, 3]

$$\text{Range} = 8 - 3 = 5$$

Limitation: Only looks at extremes; ignores distribution in between.

B. Interquartile Range (IQR) - IMPORTANT: NEWLY EMPHASIZED

Definition: Range covering the middle 50% of data; difference between third quartile and first quartile.

Quartile Definitions:

Quartile	Definition	Meaning
Q0	Minimum	Lowest value in dataset
Q1	25th percentile	25% of data points fall below this; 75% above
Q2	50th percentile (Median)	The median; middle value
Q3	75th percentile	75% of data points fall below this; 25% above
Q4	Maximum	Highest value in dataset

Formula:

$$IQR = Q3 - Q1$$

Why It Matters:

- **Robust to outliers:** Not affected by extreme values
- **Identifies outliers:** Values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$ are typically flagged as outliers
- **Shows data spread:** Wide IQR = data is spread out; narrow IQR = data is clustered

Example: If $Q1=40$ and $Q3=60$, then $IQR=20$. Middle 50% of data lies in 20-point range.

C. Variance

Definition: Measures how much data points differ from the mean.

Formula:

$$\sigma^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

Step-by-Step Example with [4, 8, 6]:

1. Calculate mean: $(4 + 8 + 6) / 3 = 6$
2. Calculate squared differences from mean:
 - $(4 - 6)^2 = 4$
 - $(8 - 6)^2 = 4$
 - $(6 - 6)^2 = 0$
3. Variance = $(4 + 4 + 0) / 3 = 2.67$

Interpretation: Larger variance = more spread out data.

D. Standard Deviation

Definition: Square root of variance; measures spread in same units as data.

Formula:

$$\sigma = \sqrt{\text{Variance}}$$

Example Continuing Above:

$$\sigma = \sqrt{2.67} \approx 1.63$$

Interpretation: Data typically varies by ± 1.63 units from mean of 6.

Beginner Hook: Variance is like measuring how far students scatter from average grade; standard deviation tells you in actual point values.

Visualization Methods for Numerical Data

A. Histograms - WITH VISUAL EXAMPLES

What It Shows: Frequency distribution of continuous data; how values are distributed.

Real Example from Slides:

- **Fare Distribution:** Shows taxi fares with most rides costing \$0-\$50, peak around \$10-\$20, right tail extending to \$500+
- **Another Example:** Shows multi-modal distribution (multiple peaks) indicating different customer segments

What You Learn:

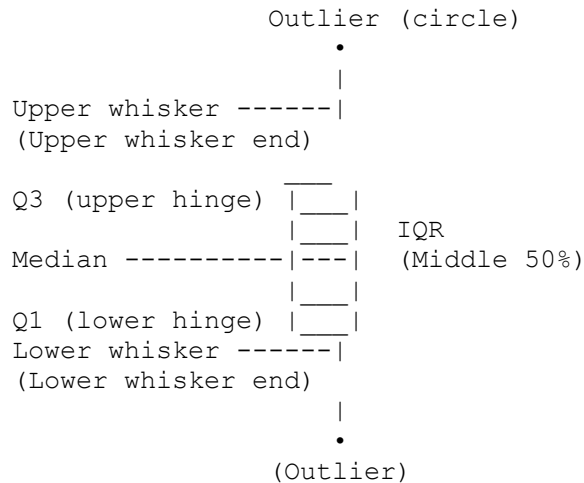
- Central tendency (where peak is)
- Spread (width of distribution)
- Shape (normal, skewed, multimodal)
- Outliers (isolated bars at extremes)

Beginner Hook: Histogram is like a bar chart of "how many values fall in each range."

B. Boxplots (Box-and-Whiskers Plot) - WITH VISUAL DIAGRAM

What It Shows: Five-point summary and potential outliers graphically.

Visual Structure (From Slide):



The Five Values:

1. **Q0 (Minimum):** Lowest value (whisker bottom)
 2. **Q1 (25th percentile):** Bottom of box
 3. **Q2 (Median, 50th percentile):** Line in box
 4. **Q3 (75th percentile):** Top of box
 5. **Q4 (Maximum):** Highest value (whisker top)
- Points beyond whiskers = outliers

Reading: Box width shows where middle 50% of data lives; position of median line shows if distribution is skewed.

Real Example from Slides:

- Taxi fare boxplots show:
 - Some fares have median around \$15
 - IQR (box height) represents typical range
 - Outliers extend to \$500+ marked as dots
 - Skew visible from median line position in box

C. Scatter Plots (and Scatter Matrix)

What It Shows: Relationship between two numerical variables.

Scatter Matrix: Matrix of scatter plots showing each pair of attributes.

Real Example from Slides:

- Switzerland heart disease dataset showing relationships between attributes

- Dense clustering in some regions indicates correlated variables
- Outlier points visible as isolated clusters

What You Learn:

- Positive correlation (slope up)
- Negative correlation (slope down)
- No correlation (random scatter)
- Outliers (isolated points)
- Clusters (groups of similar values)

D. Heatmap

What It Shows: Correlation matrix using color intensity.

Real Example from Slides:

- Shows multiple variables (rows) vs variables (columns)
- Color scale visible from blue (negative correlation) to red (positive correlation)
- White/light = no correlation
- Allows quick identification of strongly related variables

Color Interpretation:

- **Dark Red:** Strong positive correlation (variables increase together)
- **Dark Blue:** Strong negative correlation (when one increases, other decreases)
- **Light/White:** No correlation (variables independent)

Purpose: Quickly identify which variables are related without examining each scatter plot.

High-Dimensionality Visualization

Quiz Alert: Class question asks "How to visualize data in high dimensionality?"

This is for datasets with many variables (dimensions). Standard methods like scatter plots don't work well (can't plot 10 dimensions at once).

Solutions:

1. **Heatmaps:** Show correlations between all variables at once
2. **Scatter Matrices:** Show all pairwise combinations (for 5-7 variables max)
3. **Dimensionality Reduction:** PCA, t-SNE to reduce to 2-3 dimensions then visualize
4. **Parallel Coordinates Plot:** Each variable gets own axis; lines connect values
5. **Star Plots:** Each data point is star; rays represent variable values

EDA for Text Data (Natural Language Processing)

Text Analysis Goal: Provide understanding of how text is processed without human reading; derive meaningful information.

Tasks Range: Simple (word frequency) to Complex (sentiment analysis, topic modeling, translation)

Syntax vs. Semantics

A. Syntax: Structure of Language

Definition: Grammar rules; how words combine into well-formed sentences and paragraphs.

Sentence Analysis: Parse sentence into components:

- **S** = Sentence
- **N** = Noun
- **NP** = Noun Phrase
- **V** = Verb
- **VP** = Verb Phrase
- **P** = Preposition
- **PP** = Prepositional Phrase
- **ADJ** = Adjective
- **ADV** = Adverb
- **Det** = Determiner

Example: "The cat sat on the mat"

- NP: "The cat" (determiner + noun)
- VP: "sat on the mat" (verb + prepositional phrase)

B. Semantics: Meaning

Definition: Meaning of words within surrounding context; what text is actually saying.

Concept: Topical Collocation—words appearing together in certain context.

Example: "White House"

- Real estate context: compositional phrase (a white-colored building)
- Political context: topical collocation (reference to US President's residence)

Same phrase, different meaning based on context.

Beginner Hook: Syntax is grammar (Is it correct English?); Semantics is meaning (What does it actually mean?).

Text Data Pre-Processing

Before analysis, text must be transformed. Here's each technique:

1. Encoding Techniques

A. One-Hot Encoding

Purpose: Convert categorical variables into binary matrix format.

How It Works:

- Create binary column for each category
- Assign 1 if present, 0 if absent
- Each row has exactly one 1 (exclusive categories)

Example:

Original:

Item	Color
1	Red
2	Blue
3	Green
4	Red

One-Hot Encoded:

Item	Color_Red	Color_Blue	Color_Green
1	1	0	0
2	0	1	0
3	0	0	1
4	1	0	0

Quiz Alert: Know the exact format; test may provide data needing encoding.

Limitation: Creates many columns (curse of dimensionality) if variable has many categories.

B. Label Encoding

Purpose: Replace categorical values with integers (0, 1, 2, ...).

Best For: Ordinal variables where order matters.

Example:

Original:

Item	Size
1	Small
2	Medium
3	Large
4	Small

Label Encoded:

Item	Size
1	0
2	1
3	2
4	0

Why It Works for Ordinal Data: Small (0) < Medium (1) < Large (2) maintains hierarchy.

Warning: Don't use for non-ordinal categorical data (colors don't have order).

2. Case Normalization

Definition: Converting all words to either uppercase or lowercase.

Function: `lower()` function performs conversion.

Important Note: Not always necessary!

When to Do It:

- Information retrieval tasks (search)
- When case shouldn't matter

When NOT to Do It (Keep Original Case):

- Named Entity Recognition (names should stay capitalized)
- When word meaning changes with case

Example of Meaning Change:

- "polish" (language) vs "Polish" (nationality)
- "May" (month) vs "may" (possibility)

Beginner Hook: Don't blindly lowercase—case carries meaning sometimes.

3. Removing Stop Words

Stop Words Definition: Extremely common words carrying little lexical content.

Examples:

- Articles: "a", "the", "an"
- Pronouns: "he", "him", "they", "I"
- Particles: "well", "however", "thus"

How Determined: Compute total appearances in corpus; most frequent = stop words.

Why Remove:

- Don't distinguish documents from each other
- Presence doesn't help with analysis
- Skew analysis results if kept
- Reduce noise in text analysis

Example:

- Email analysis: Remove headers ("Subject", "To", "From"), legal disclaimers
- Sentiment analysis: "the" and "and" tell you nothing about sentiment

Caution: Sometimes stop words matter (double negatives in sentiment: "not bad" = positive).

4. Stemming

Definition: Remove prefixes, suffixes, and pluralization, leaving root stem.

Goal: Map multiple word forms to single stem.

Examples:

watches → watch
parties → party
carrying → carry
running → run

Method: Rules-based; applies known rules without context.

Limitation: May produce non-words; doesn't always give linguistically correct root.

Speed Advantage: Fast; no dictionary lookups needed.

5. Lemmatization

Definition: Advanced form of stemming using context, vocabulary, morphology, and grammar to find lemma (dictionary form).

How Different from Stemming:

- Uses grammatical information (part-of-speech tags)
- Considers surrounding context
- Returns actual dictionary words

Examples:

better → good (understands "better" is comparative of "good")
was → be (identifies verb conjugation)
running → run (knows it's present participle)

Advantage: More linguistically accurate; produces real words.

Trade-off: Slower than stemming; needs dictionary and grammar information.

Beginner Hook: Stemming chops carelessly (watches→watch, carrying→carr); lemmatization intelligently conjugates (was→be, better→good).

6. Text Vectorization - Count Vectorizer (DETAILED METHOD)

Purpose: Convert collection of text documents into numerical matrix—words become features, documents become rows, values are word counts.

Three-Step Process:

Step 1: Tokenization

- Split text into individual tokens (usually words)
- Remove punctuation
- Convert to lowercase

Step 2: Vocabulary Building

- Create list of all unique tokens across entire corpus
- Each unique word gets an index/column

Step 3: Count Calculation

- For each document, count frequency of each word
- Create vector where each dimension = word count in vocabulary

COMPLETE EXAMPLE:

Input Documents:

Doc 1: "The cat sat on the mat."

Doc 2: "The dog sat on the log."

Step 1 (Tokenization):

Doc 1: ["the", "cat", "sat", "on", "the", "mat"]

Doc 2: ["the", "dog", "sat", "on", "the", "log"]

Step 2 (Vocabulary):

["the", "cat", "sat", "on", "mat", "dog", "log"]
(7 unique words)

Step 3 (Count Matrix):

	the	cat	sat	on	mat	dog	log
Doc 1	2	1	1	1	1	0	0
Doc 2	2	0	1	1	0	1	1

Interpretation:

- Doc 1 has word "the" appearing 2 times, "cat" 1 time, etc.
- Doc 2 has word "the" appearing 2 times, "cat" 0 times, etc.
- Diagonal patterns show unique vocabulary per document

Output Use: This numerical matrix can be fed to machine learning algorithms.

PART 4: REGULAR EXPRESSIONS (REGEX)

Regular Expression Definition: A set of symbols that describes a text pattern.

Why Regular Expressions Matter:

For Finding & Replacing:

- Search document for "colour" or "neighbor" (with/without 'u')

- Convert tab-delimited file to comma-delimited file
- Find duplicate words in text
- Search and replace "Bob" and "Bobby" with "Robert"

For Validation:

- Phone numbers: Does it have right number of digits?
- Emails: Valid format?
- Dates: Correct format? Month ≤ 12 ?

Beginner Hook: Regex is like playing "Guess my pattern"—you give rules describing what you're looking for, and it finds everything matching those rules.

Character Sets

[...] notation matches any ONE character from the set inside brackets.

Basic Character Sets

Matching Alternatives:

Pattern: `gr[ea]y`

- Matches: "grey", "gray"
- Does NOT match: "grAy", "graay", "graey"

Key Point: Only ONE character from the set is matched. Sets are order-independent.

Quiz Alert: Test will ask what this pattern matches/doesn't match.

Character Ranges

[a-z], [A-Z], [0-9] notation

Instead of listing every character, use ranges:

- `[a-z]`: lowercase letters a through z
- `[A-Z]`: uppercase letters A through Z
- `[0-9]`: digits 0 through 9
- `[a-zA-Z]`: any letter
- `[a-zA-Z0-9]`: any letter or digit

Example: Victorian Car Plate Pattern - VISUAL BREAKDOWN

Plates look like: XRA 000, 1AA 1AA

Pattern with Visual Breakdown:

```
[A-Z0-9] [A-Z] [A-Z] \s [0-9] [A-Z0-9] [A-Z0-9]
```

[A-Z0-9] = One of: "A"-"Z" OR "0"-"9" (FIRST CHAR)

[A-Z] = One of: "A"-"Z" (SECOND CHAR)

[A-Z] = One of: "A"-"Z" (THIRD CHAR)

\s = whitespace (SPACE)

[0-9] = One of: "0"-"9" (FOURTH CHAR)

[A-Z0-9] = One of: "A"-"Z" OR "0"-"9" (FIFTH CHAR)

[A-Z0-9] = One of: "A"-"Z" OR "0"-"9" (SIXTH CHAR)

Matching Examples:

- ✓ XRA 000
- ✓ 1AA 1AA
- ✓ ABC 999
- ✗ xra 000 (lowercase x)
- ✗ AB 000 (only 2 letters, needs 3)

Quiz Alert: CAUTION—[50-99] does NOT mean 50 to 99. It means "5", "0", "9", "9" (individual characters). For number ranges, you need different approach.

Negative Character Sets

[^...] notation matches any character NOT in the set.

Example:

[^b]og matches "hog" and "dog" but NOT "bog"

Interpretation: "Any character except 'b', followed by 'og'"

Caution Questions:

- Does `see[^mn]` match "see"? **NO** (needs a character after)
- Does `see[^mn]` match "see "? **YES** (space is not m or n)

Beginner Hook: [^] is the opposite of []—instead of "one of these", it's "anything except these".

Metacharacters Inside Character Sets

Special Rule: Metacharacters LOSE their special meaning inside [...].

Example:

`h[ai.u]t` matches "hat", "hit", "hut", "h.t"
The period is literal, not "any character"

Exceptions That Still Need Escaping:

- `]` (closes bracket)
- `-` (indicates range)
- `^` (indicates negation if first)
- `\` (escape character)

Example Pattern:

```
var[([0-9])]
```

Matches: `var(0, var(1, ..., var(9, var[0], var[1], ..., var[9]`

Shorthand Character Sets

To avoid writing long character sets, use shortcuts:

Shorthand	Meaning	Equivalent To
<code>\d</code>	Any decimal digit	<code>[0-9]</code>
<code>\w</code>	Any word character	<code>[a-zA-Z0-9_]</code>
<code>\s</code>	Any whitespace	<code>[\t\n\r]</code> (space, tab, newline, return)
<code>\D</code>	Any NON-digit	<code>[^0-9]</code>
<code>\W</code>	Any non-word character	<code>[^a-zA-Z0-9_]</code>
<code>\S</code>	Any non-whitespace	<code>[^ \t\n\r]</code>

Examples:

`\d\d\d\d` matches "2016" (four digits), not "abcd"
`\w\w\w` matches "abc", "123", "d_b" (three word characters)
`\w\w\s\w` matches "ab c" (two words, space, one word), not "a bc"
`[\\w]-[\\w]` matches "a-b", "1-2" (char-char separated by hyphen)

Critical Caution:

`[^\d\s]` is NOT the same as `[^\D\S]`

`[^\d\s]`: "Neither a digit NOR a space" `[^\D\S]`: "Either a non-digit OR a non-space" (very different!)

Example:

- `[^\d\s]` matches letter "a" (✓ not digit, ✓ not space)
- `[^\D\S]` matches "a" (✓ it's non-digit) AND space (✓ it's non-space)

Quiz Alert: Know this distinction—test specifically asks about it.

Repetition Meta-Characters

Meta-character	Meaning	Example
*	0 or more repetitions of preceding regex	<code>ab*c</code> matches "ac", "abc", "abbc", "abbbc"
+	1 or more repetitions	<code>ab+c</code> matches "abc", "abbc" (NOT "ac")
?	0 or 1 repetition	<code>ab?c</code> matches "ac", "abc" (NOT "abbc")

Example Test Question:

Match: oops, oops, oooooops, oooooops NOT: ops

Which patterns work?

- A. `oo*ps` ✗ (could match "ops")
- B. `ooo*ps` ✗ (would need "ooo" minimum)
- C. `oo+ps` ✓ (needs "oo" then 1+ "o"s then "ps")
- D. `oo?ps` ✗ (would match "ops", "oops" only)

Answer: C — `oo+ps` requires at least 2 o's.

Quantified Repetitions

{m,n} syntax: Match exactly from m to n repetitions.

Format:

- `\d{2}` = exactly 2 digits
- `\d{2,4}` = 2 to 4 digits
- `\d{2,}` = 2 or more digits (n infinite)

Rules:

- m (minimum) must always be included, can be 0
- n (maximum) is optional
- Both are positive numbers

Example Pattern:

Match: report_2016_09, assignment_2016_9, budget_16_08, assignment_08_7

NOT: report_201609_39, assignment_6_9000, budget_2345678_08

Pattern:

`\w+_ \d{2,4}_ \d{1,2}`

Breakdown:

- `\w+`: One or more word characters
- `_`: Literal underscore
- `\d{2,4}`: 2 to 4 digits (allows "16" through "2016")
- `_`: Literal underscore
- `\d{1,2}`: 1 to 2 digits (allows "7" or "09")

Grouping (...) - VISUAL DIAGRAMS FROM SLIDES

Parentheses (...) serve four purposes:

1. **Apply repetition to groups of patterns** (not just single characters)
2. **Make regex easier to read**
3. **Capture groups for extraction** (retrieve matched content)
4. **Cannot use inside character sets**

Grouping Examples:

Example 1: Repetition

`abc+` matches "abc", "abcc", "abccccc" (repeats "c" only)

`(abc)+` matches "abc", "abcabc", "abcabcabc" (repeats whole group)

Visual Breakdown from Slide:

abc+
"a" "b" "c" (+ applies only to c)

(abc)+
"abc" (+ applies to whole group)

Example 2: Text Extraction

Text: "Incident American Airlines Flight 11 involving a Boeing 767-223ER in 2001"

Pattern: (.*?) involving

Capture group 1 = "Incident American Airlines Flight 11"

Visual Breakdown from Slide:

"Incident "
(any character)
" involving"
↓
GROUP #1 captures everything before "involving"

Beginner Hook: Parentheses let you "grab" what matched and use it later.

Alternation (|)

"|" is the OR operator

- A|B matches strings matching either A OR B
- Order matters: leftmost gets precedence
- Can chain multiple: A|B|C|D
- Must group to keep distinct: (apple|orange) juice

Examples:

apple|orange matches "apple" and "orange"

(apple|orange) juice matches "apple juice" and "orange juice"
NOT "applejuice" or "apple orangejuice"

w(ei|ie)rd matches "weird" and "wierd"

The Backslash Plague

Backslash \ serves two functions:

1. Indicates special forms (e.g., \d, \s)
2. Escapes special characters to use literally

The Problem:

To match literal backslash:

Text string to match: \section

Regular expression string: \\section
(Each pair of \\ becomes single \)

Python string literal: r"\\section"
(Use raw string r"..." to avoid confusion)

Solution: Raw Strings

Raw string r"..." suppresses escape character interpretation.

Regular Python String Raw String Equivalent

```
"\\\\\\\\section"    r"\\\\section"  
"\\\\w+\\\\s+"        r"\\w+\\s+"
```

Best Practice: Always use raw strings for regex patterns in Python.

```
import re  
pattern = r"\d{2,4}_\d{1,2}" # Much cleaner than "\\d{2,4}_\\d{1,2}"
```

Quiz Alert: Test may ask about escaping and raw strings—know the difference.

Email Validation Example

Real-world regex for email validation:

```
r"^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$"
```

Pattern Breakdown:

- ^ = start of string
- [a-zA-Z0-9_.+-]+ = one or more word chars, dot, plus, or hyphen (local part)
- @ = literal @
- [a-zA-Z0-9-]+ = one or more word chars or hyphen (domain name)
- \. = literal dot (escaped because . normally means "any char")

- [a-zA-Z0-9-.-]+ = one or more word chars, dot, or hyphen (TLD)
 - \$ = end of string
-

PART 5: DATA DISCOVERY & COLLECTION IN DETAIL

Data Discovery Definition & Purpose

Data Discovery is the process of identifying and understanding data sources that can be used for analytical purposes.

Primary Purposes:

- Gain actionable insights into available data
- Understand potential for analysis
- Determine how it supports decision-making and research objectives

Beginner Hook: Data discovery is detective work—finding what data exists, where it lives, and whether it's useful for your investigation.

Challenges in Data Discovery

Eight Major Challenges:

1. **Volume and Complexity of Data** - Too much data from too many sources
 2. **Data Quality and Silos** - Good data scattered; poor quality data isolated
 3. **Dynamic and Evolving Data** - Data changes constantly; metadata becomes outdated
 4. **Data Privacy and Security Concerns** - Can't access sensitive data; legal restrictions
 5. **Lack of Metadata and Documentation** - Don't know what data exists or what it means
 6. **Interoperability and Integration Issues** - Different systems, formats; hard to combine
 7. **Resource Constraints** - Limited budget, tools, expertise, time
 8. **Finding Actionable Insights** - Knowing what's available doesn't mean finding useful data
-

PART 6: DATA COLLECTION IN DETAIL

Data Collection Definition & Characteristics

Data Collection is the systematic process of gathering and measuring information on variables of interest in an established systematic fashion enabling research questions to be answered, hypotheses tested, outcomes evaluated.

Key Characteristics:

- **Systematic:** Follows documented process
 - **Intentional:** Targets specific variables
 - **Standardized:** Uses consistent methods
 - **Purposeful:** Answers specific questions
-

Primary vs. Secondary Data

Primary Data (Original, Collected Firsthand)

- Researcher controls quality, methodology, scope
- Expensive and time-consuming
- Specific to research purpose
- Full understanding of collection process

Secondary Data (Already Collected)

- Cost and time efficient
- Access to broad, diverse datasets
- May not match current needs perfectly
- Quality varies; may be outdated

Beginner Hook: Primary = doing your own experiment; Secondary = reading someone else's research results.

Quantitative vs. Qualitative Data

Dimension	Quantitative	Qualitative
Nature	Numerical, measurable	Descriptive, conceptual
What Measured	Quantity and magnitude	Qualities and characteristics
Collection	Surveys, experiments, structured instruments	Interviews, observation, open-ended questions
Analysis	Statistical hypothesis testing	Interpretive, thematic analysis
Display	Graphs, charts, tables	Text, transcripts, narratives
Purpose	Formulate facts, patterns	Understand concepts, generate hypotheses

Sample Size	Large samples	Small, purposive samples
--------------------	---------------	--------------------------

Quiz Alert: Know comparison table—test will ask differences.

Data Collection Methods for Structured Data

Method 1: Surveys and Questionnaires

Key Considerations:

1. **Clearly Define Objectives** - What do you want to learn?
2. **Question Design** - Clear, unbiased, appropriate question types
3. **Sampling** - Representative sample selection
4. **Pilot Testing** - Test instruments before full rollout
5. **Ethical Considerations** - Informed consent, avoid harm
6. **Distribution Method** - Paper, online, telephone, face-to-face?

Advantages: Structured, comparable responses; statistical analysis possible

Disadvantages: Response bias; shallow understanding; time-consuming design

Method 2: Online Forms

Design Considerations:

- **User Experience & Design** - Easy navigation, clear instructions
 - **Privacy & Security** - SSL encryption, data protection
 - **Accessibility** - Mobile-friendly, screen reader compatible
 - **Data Quality** - Input validation, required fields, reasonable ranges
-

Method 3: Web Scraping

Definition: Automated extraction of data from websites.

Key Considerations:

- **Legal & Ethical** - Respect robots.txt, terms of service
- **Technical Challenges** - JavaScript rendering, changing page structure
- **Data Quality & Relevance** - Ensure extracted data is accurate and useful
- **Efficiency & Resources** - Manage bandwidth; avoid overloading servers

Caution: Always check legal permissions before scraping.

Method 4: Relational Databases

Design Considerations:

- **Database Design & Structure** - Logical schema supporting efficient queries
 - **Data Quality & Integrity** - Constraints, validations, referential integrity
 - **Scalability & Performance** - Indexing, partitioning for growth
 - **Security Measures** - Access control, encryption, audit logs
 - **Backup & Recovery** - Durability and disaster recovery
 - **Compliance** - Meet GDPR, HIPAA, SOX requirements
 - **Data Accessibility & Documentation** - Clear metadata; easy access
 - **Monitoring & Maintenance** - Performance optimization; proactive monitoring
-

Method 5: APIs (Application Programming Interfaces)

Key Considerations:

- **API Documentation Review** - Understand endpoints, parameters, limitations
- **Authentication & Authorization** - API keys, OAuth, secure access
- **Rate Limiting & Quotas** - Respect server limits; implement backoff
- **Data Pagination** - Handle large result sets efficiently
- **Error Handling** - Manage timeouts, retries, error codes
- **Data Efficiency & Minimization** - Request only needed fields
- **API Terms of Service Compliance** - Follow usage agreements
- **Data Storage & Management** - Handle API responses appropriately
- **Monitoring & Maintenance** - Track API changes; monitor costs

Example: Twitter API, Google Maps API, OpenWeather API

Data Collection Methods for Unstructured Data

Method 1: Text Mining and NLP

Key Considerations:

- **Data Quality & Volume** - Enough quality text?
- **Data Preparation & Pre-processing** - Tokenization, stemming, lemmatization
- **Choose Right NLP Techniques** - Sentiment analysis, topic modeling, NER?

- **Understanding Context & Nuances** - Sarcasm, metaphor, cultural references
 - **Ethical Considerations & Bias** - Avoid biased models, fair representation
 - **Performance Evaluation & Validation** - Test quality; validate results
 - **Scalability & Resources** - Computational requirements for large corpora
 - **Integration** - Combine with other data sources
-

Method 2: Image and Video Data Collection

Key Considerations:

- **Consistent Quality & High-Resolution** - Ensure consistent image quality
 - **Diversity & Representation** - Cover different scenarios, demographics
 - **Ethical Considerations & Legal Compliance** - Privacy, consent, copyright
 - **Large File Sizes & Data Organization** - Storage and labeling challenges
 - **Accurate Annotations & Labeling** - Critical for supervised learning
 - **Conversion & Processing** - Standardize formats for pipelines
 - **Ethical Use & Bias Mitigation** - Avoid perpetuating biases
 - **Bandwidth & Transfer** - Slow transfer speeds for large video files
 - **Real-Time Processing** - Handle streaming video if applicable
-

Method 3: Social Media and Web Content

Key Considerations:

- **Legal & Ethical Considerations** - Terms of service, privacy laws
 - **Adherence to APIs Terms of Use** - Follow rate limits, attribution requirements
 - **Changes in API Access** - APIs change; old access may break
 - **Dynamic Content** - Content changes constantly
 - **Handling Noise** - Spam, bots, irrelevant content
 - **Anonymization & Data Processing** - Protect user privacy
 - **Sampling Bias & Cultural Context** - Social media doesn't represent all populations
 - **Long-Term Accessibility** - Content deleted; links break
 - **Archiving & Preservation** - Plan for long-term storage
-

Data Collection Methods for Semi-Structured Data

Method 1: JSON and XML Data Extraction

Data Structures:

- JSON (JavaScript Object Notation): Lightweight, key-value pairs
- XML (Extensible Markup Language): Hierarchical, self-describing tags

Key Considerations:

- **Understand Data Structure** - Hierarchical? Schema-based?
 - **Parsing Data Using Libraries** - Python: json, xml.etree; appropriate libraries
 - **Regular Expressions** - Extract specific patterns from text
 - **Character Encoding** - Proper UTF-8 handling
 - **Handling Inconsistencies & Errors** - Malformed data; error recovery
 - **Schema Validation** - Verify data matches expected structure
-

Method 2: Logs and Sensor Data Collection

Characteristics: High-volume, continuous streams from devices/systems.

Key Considerations:

- **Volume & Velocity** - High throughput requires stream processing
 - **Variability & Structure** - Diverse formats; standardization needed
 - **Time-Sensitivity** - Timestamps and timezone awareness critical
 - **Interoperability** - Different log formats from different systems
 - **Automated Alerts & Actions** - Real-time response to anomalies
-

Method 3: Email and Communication Data Collection

Key Considerations:

- **Privacy & Legal Compliance** - GDPR, consent, sensitive information
 - **Data Structuring & Formatting** - Complex structures, nested attachments
 - **Data Quality & Integrity** - Deduplication, noise filtering
 - **Metadata Extraction** - Sender, date, recipients crucial
 - **Handling Attachments** - Extract, classify, analyze separately
-

Ethical Considerations and Privacy

Eight Ethical Principles for Data Collection:

1. Informed Consent

- Transparency and openness about data use
- Voluntariness (no coercion)
- Participants understand what they're consenting to

2. Privacy and Anonymity

- Protect personal information
- Anonymize when possible
- Pseudonymization where full anonymity impossible

3. Data Security

- Secure storage and transmission
- Access controls (least privilege)
- Encryption of sensitive data

4. Compliance with Laws & Regulations

- GDPR (European Union)
- HIPAA (Healthcare, US)
- Local data protection laws

5. Data Retention & Disposal

- Clear retention policies
- Secure, complete deletion when policy met

6. Respect for Participants

- Respect autonomy (choices honored)
- Beneficence (maximize benefits) and non-maleficence (minimize harm)
- Accountability for data use

7. Equity & Fairness

- Inclusive data collection (represent all populations)
- Fair treatment regardless of demographic

8. Data Minimization & Necessity

- Collect only necessary data
- Limit use to stated purpose

Quiz Alert: Know all eight principles—heavily tested concept.

Importance of Data in Decision-Making

Data-Driven Decision-Making Powers:

Area	Data's Role
Evidence-Based Decisions	Objective analysis; insight generation; trend analysis
Strategic Planning	Resource allocation; competitive benchmarking; roadmap development
Risk Management	Identifying risks; assessing impact; mitigation strategies
Performance Optimization	Operational efficiency; benchmarking; continuous improvement
Customer Insights	Understanding behavior; personalization; service improvement
Innovation	Identifying market needs; feedback loops; product development
Regulatory Compliance	Compliance monitoring; transparency; reporting
Crisis Management	Crisis prediction; rapid response; building resilience

In Research: Data enables quantitative analysis, comparative studies, hypothesis testing, theory development, informing policy, innovation, and ethical/social responsibility.

FINAL SUMMARY TABLE: THE COMPLETE DATA WRANGLING WORKFLOW

Phase	Goal	Key Tasks	Output	Tools
Discovery	Identify usable data sources	Inventory existing data; profile; assess quality	Data catalog; metadata	Data discovery tools
Collection	Gather required data systematically	Design collection; execute; document	Raw data	APIs, databases, forms, scraping
Cleaning	Remove errors and inconsistencies	Identify anomalies; correct/remove; validate	Clean dataset	Python (Pandas), regex
Pre-processing	Prepare data for analysis	Select, format, standardize	Standardized data	Python, R
Transformation	Convert to analysis-ready format	Normalize, aggregate, encode, vectorize	Transformed data	Python (Pandas, scikit-learn)
Enrichment	Add context and depth	Integrate sources; augment records; add semantic info	Enriched dataset	Database joins, APIs

Storing	Save for access and analysis	Design schema; implement security; optimize	Stored data	Databases, warehouses, cloud
Validation	Ensure quality and usability	Define rules; test accuracy; check completeness	Validated data	Python validation scripts

QUICK REFERENCE: KEY DEFINITIONS

Data Wrangling: Process of acquiring, cleaning, structuring, and enriching raw data into format directly usable for analysis.

Data Discovery: Process of identifying and understanding data sources for analytical purposes.

Data Collection: Systematic gathering and measuring of variables of interest.

EDA: Exploratory Data Analysis—exploration of data characteristics using statistics and visualizations.

Regular Expression: Set of symbols describing text pattern for finding, replacing, validating.

Tokenization: Breaking text into individual units (words, phrases, characters).

Lemmatization: Reducing words to dictionary form using grammar and context.

Stemming: Removing prefixes/suffixes to get word root.

Metadata: Information about data itself (origin, format, characteristics).

Normalization (Data): Rescaling data to 0-1 range; also organizing database to reduce redundancy.

Standardization: Adjusting data to mean 0, standard deviation 1.

One-Hot Encoding: Converting categorical variables to binary matrix.

Label Encoding: Replacing categories with integers.

Interquartile Range (IQR): Range covering middle 50% of data; Q3 - Q1.

Boxplot: Five-point summary showing min, Q1, median, Q3, max, and outliers.

QUIZ PREPARATION CHECKLIST

High-Probability Quiz Questions:

□ Define data wrangling and its eight core tasks □ List eight challenges in data wrangling □ Explain why data wrangling takes 60% of data scientist time □ Describe complete data discovery process with six components □ Distinguish primary vs. secondary data □ Compare quantitative vs. qualitative data collection □ Explain six types of metadata □ Know five reasons for understanding data types in EDA □ Calculate mean, median, mode, variance, standard deviation, IQR □ Describe IQR formula and quartile definitions (Q0-Q4) □ Describe four visualization methods (histogram, boxplot, scatter, heatmap) with visual examples □ Explain NLP concepts: syntax vs. semantics □ Know five text preprocessing techniques and when to use each □ Distinguish between stemming and lemmatization □ Understand Count Vectorizer three-step process with example □ Know all regex shorthand characters and their meanings □ Identify correct regex patterns from examples (the "oops" question always appears) □ Explain grouping, alternation, and quantified repetitions in regex with visual breakdowns □ Distinguish `[\d\s]` vs `[D\S]` □ List all twelve data collection considerations □ Recall eight ethical principles for data collection □ Match data transformation methods to data types □ Identify continuous vs. discrete data with examples □ Understand boxplot structure with five-point summary □ Read heatmaps and understand color intensity meanings □ Know ASCII table basics for character data

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